Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 1 of 11 · Question 2 part c · 2 marks

2025 JUN · Paper 02 (regional R) · find part c in this mark scheme extract

2	(a)	**C (endoplasmic reticulum)** The answer is not A as the cell membrane transports substance into and out of the cell B cytoplasm as this is a base for chemical reactions D as ribosomes provide sites for protein synthesis		1
	(b) (i)	Z Y W X;		1
	(ii)	growth/repair;	Ignore repair of body cells	1
	(c)	Any two from • mitosis produces diploid cells/46 chromosomes, meiosis produces haploid cells/23 chromosomes; • mitosis produces 2 (daughter) cells, meiosis produces 4 cells/gametes; • (daughter) cells from mitosis are genetically identical, (daughter) cells from meiosis show variation;		2
			Total for Question 2 = 5 marks	

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 2 of 11 · Question 5 part a.ii · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number					
5	(a)	(i)	A (DNA); B/C/ and D are found in or associated with other structures.		1
		(ii)	correct circles of pair;		1
		(iii)	only two pairs of homologous chromosomes/ would be 23 in a human cell;		1
	(b)		• prophase, metaphase, anaphase and telophase; • (prophase) chromosomes become thicker/visible; • (metaphase) chromatids line up at equator; • (anaphase) chromatids separate and migrate to poles; • (telophase) nucleus/nuclear membrane reforms;	Allow condense Allow chromosomes throughout Allow clear annotated diagrams showing stages	5

Total 8 marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 2 of 11 · Question 5 part a.ii · 1 mark

2021 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 3 of 11 · Question 5 part a.iii · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

Question number					
5	(a)	(i)	A (DNA); B/C/ and D are found in or associated with other structures.		1
		(ii)	correct circles of pair;		1
		(iii)	only two pairs of homologous chromosomes/ would be 23 in a human cell;		1
	(b)		• prophase, metaphase, anaphase and telophase; • (prophase) chromosomes become thicker/visible; • (metaphase) chromatids line up at equator; • (anaphase) chromatids separate and migrate to poles; • (telophase) nucleus/nuclear membrane reforms;	Allow condense Allow chromosomes throughout Allow clear annotated diagrams showing stages	5

Total 8 marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 3 of 11 · Question 5 part a.iii · 1 mark

2021 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 4 of 11 · Question 5 part c · 4 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

number				
5	(a)	• change in base sequence; • causes change in protein synthesis; • causes change in phenotype;		3
	(b) (i)	• axes labelled with units; • suitable scale; • correct plots; (3 x 2) • bars of equal width;		5
	(ii)	• 43 000 × 0.0015; • = 64.5; • 65;		3
	(c)	• extra chromosome at pair 21; • diploid number 47; • female because 2 X chromosomes; • at pair 23;		4

Total : 15 marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 4 of 11 · Question 5 part c · 4 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 5 of 11 · Question 3 part b.ii · 1 mark

2023 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

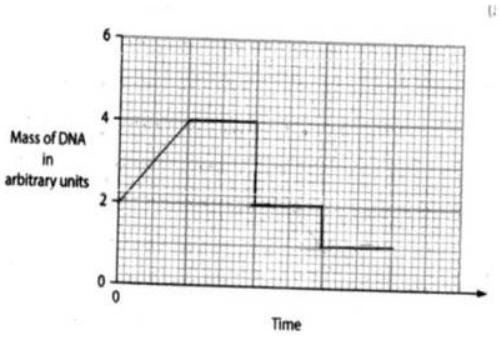
3 (a)	• contains particular features / characteristics/ adaptations; • to perform a specific function;		2
(b) (i)	• 84 ÷ 0.006; • 14000x	Allow correct final answer from a calculation that includes 84 and 0.006, max 1 mark	2
(ii)	half the amount/haploid/23 chromosomes;		1
(iii)	• to provide energy; • for swimming/movement/travelling;		2
(iv)	• 180 ÷ 0.070; • 2571; • (÷ 60) 42.85;	Allow any figure that rounds to 43 Allow ecf for third mark - an incorrect value for a calculation carried out for mps1 and 2, divided by 60 **and** a correct answer to that calculation	3 exp
(c)	Any two from: • greater magnification; • higher resolution/can see objects more clearly/in more detail; • can see a greater number of structures;		2 exp

Total Question 3 = 12 marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 6 of 11 · Question 4 part b.iii · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
4	(a) (i)	**A; (chromosomes)** B/C/D do not contain DNA		1 1
	(ii)	RNA/mRNA/tRNA;		
	(iii)	<table><tr><th>Differences</th><th>DNA</th><th>Named nucleic acid</th></tr><tr><td>1</td><td>double stranded</td><td>single stranded</td></tr><tr><td>2</td><td>contains deoxyribose</td><td>contains ribose</td></tr><tr><td>3</td><td>contains thymine</td><td>contains uracil</td></tr></table>		3
	(b) (i)	• replication; • formation of a new strand (of DNA); • by complementary base pairing; • double the original mass of DNA;		4
	(ii)	• cell dividing; • cytokinesis;		2
	(iii)	• line starting at two; • increases to four; • ends at one;  		3

Total 14 marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 6 of 11 · Question 4 part b.iii · 3 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 7 of 11 · Question 8 part d · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part d in this mark scheme extract

8 (a)	Three from: • (parasite) carried by mosquito/mosquito is a vector; • mosquito bites human; • parasite/infected blood drawn/sucked up into mosquito; • infected blood transmitted to other people;		Max 3
(b)(i)	• fewer deaths caused by *P.vivax* in R than Q • (but) more deaths caused by *P.vivax* in R than in Q as a proportion of the total; • 190 000 fewer deaths caused by *P.vivax* in R/300 less deaths in Q caused by *P.vivax*; • 0.5% deaths caused by *P.vivax* in Q/62.5% of deaths caused by *P.vivax* in R;		1 1 1
(ii)	more mosquitoes in one region than another/climate favours breeding of mosquitoes/more dense population of people/better health care/use of (named) preventative measures;		1
(c)	Three from: Fewer people with malaria/reduced incidence of malaria; Immunity/antibodies against parasite/herd immunity; Reduction in population of/death of parasite; Less transmission (from one person to another);		Max 3
(d)	Three from: (sexual reproduction) • gives rise to variation in offspring; • variation provides a survival advantage; • parasite more likely to survive in a changing environment; (asexual reproduction) • parasite can reproduce faster; • no need to find a mate/only one parent needed; • larger number of offspring produced;	Answer must contain at least one advantage of each method	Max 3

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 7 of 11 · Question 8 part d · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 8 of 11 · Question 8 part d · 4 marks

2020 NOV · Paper 01 · find part d in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 9 of 11 · Question 6 part b.ii · 4 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
6	(a)	(i)	• vas deferens cut; • sperms can't pass/not released (to outside); • don't reach vagina/uterus/ovum; • fertilisation can't occur;		4
		(ii)	• fluid still passed; • from prostate gland; • this can contain HIV;		3
	(b)	(i)	X on testes;		1
		(ii)	• sperm and ova/male and female gametes; • both have haploid/(n) chromosomes; • fuse at fertilisation; • diploid (2n) number restored;		4

Total 12 marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 9 of 11 · Question 6 part b.ii · 4 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 10 of 11 · Question 2 part a.iii · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number					
2	(a)	(i)	movement/swimming;		1
		(ii)	C; A, B and D do not release energy		1
		(iii)	three from: • produces a haploid cell / reduces the number of chromosomes by half/to 23; • so at fertilisation the diploid/full number of/46 chromosomes is restored; • prevents doubling of chromosomes at fertilization; • introduces (genetic) variation;		3
	(b)	(i)	sperm egg X / Y X / XX / XY X / XX / XY ;	One for egg One for sperm One for the offspring	3
		(ii)	Two from: • acrosome; • contains enzymes; • digests/breaks down the egg cell membrane;		2

Total 10 marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 10 of 11 · Question 2 part a.iii · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare haploid and diploid cells and explain why meiosis matters

Answer 11 of 11 · Question 4 part b.ii · 4 marks

2024 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

4 (a)	(i)	11, 31, 30		(1)
	(ii)	31 (× 100); 100 = 31%;	send to review if other than 31	(2)
(b)	(i)	**D (alternative forms of the same gene in a genotype)** The following are incorrect because: A two dominant genes are the same type of allele B two recessive genes are the same type of allele C alleles are not different genes		(1)
	(ii)	• sperm and egg/gametes each contain one allele/haploid; • sperm and egg/gametes fuse; • during fertilisation; • (fertilisation) brings alleles together (in diploid cell)/chromosomes (from each gamete) are paired/one allele comes from each parent;		(4)
(c)	(i)	Hh/heterozygous; hh/homozygous recessive;	either order any suitable letters to be used for alleles	(2)
	(ii)	wheezing/coughing/shortness of breath/low weight / fatigue/over production of mucus/diarrhoea/male infertility;	allow other valid symptoms	(1)

Total for Question 4 = 11 marks